

EventEase — Entity Relationship Diagram

Diagram

Entities

Venue

Field	Type	Constraints	Description
VenueId	int	PK, auto-increment	Unique venue identifier
VenueName	string	Required, max 100, Unique	Name of the venue
Location	string	Required, max 200	Physical address or area
Capacity	int	Required, range 1–100,000	Maximum number of attendees
ImageUrl	string?	Optional, max 500	Path to venue image or external URL

Event

Field	Type	Constraints	Description
EventId	int	PK, auto-increment	Unique event identifier
EventName	string	Required, max 150	Name of the event
Description	string?	Optional, max 500	Details about the event
VenueId	int	FK → Venue, Required	Venue where the event is held

Booking

Field	Type	Constraints	Description
BookingId	int	PK, auto-increment	Unique booking identifier
EventId	int	FK → Event, Required	Event being booked
VenueId	int	FK → Venue, Required	Venue being booked
EventDate	DateTime	Required	Date of the booking
StartTime	TimeSpan	Required	Start time of the booking
EndTime	TimeSpan	Required	End time of the booking

Relationships

Relationship	Cardinality	FK Column	Delete Behavior
Venue → Event	One-to-Many	Event.VenueId	Restrict
Venue → Booking	One-to-Many	Booking.VenueId	Restrict
Event → Booking	One-to-Many	Booking.EventId	Restrict

Delete Behavior: Restrict — A venue cannot be deleted if it has associated events or bookings. An event cannot be deleted if it has associated bookings. This is enforced at both the database (FK constraint) and application (controller) level.

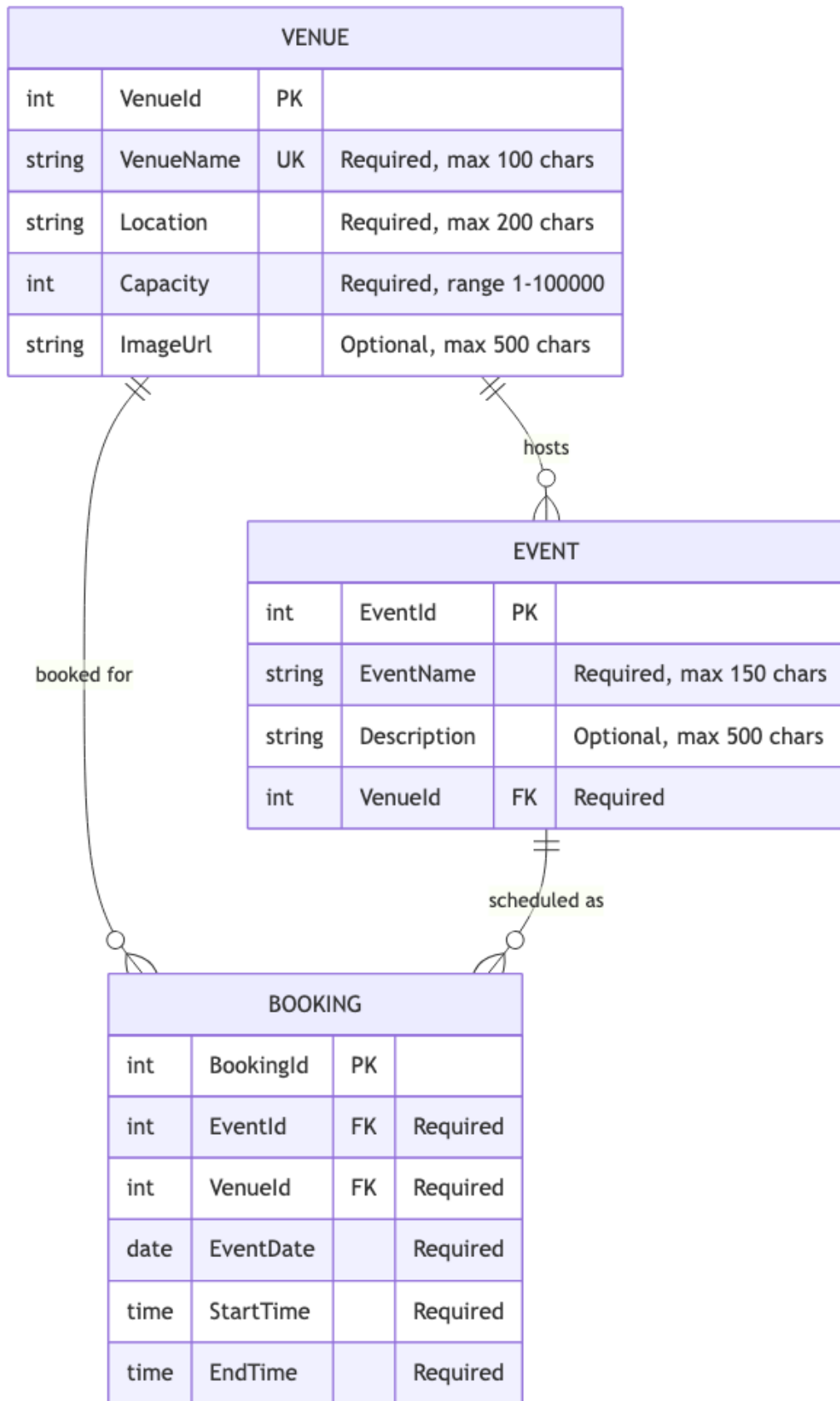


Figure 1: ERD Diagram

Business Rules

1. **Double-booking prevention:** A booking cannot be created if another booking exists for the same venue on the same date with overlapping time ($\text{StartTime} < \text{other.EndTime}$ AND $\text{EndTime} > \text{other.StartTime}$).
2. **Venue deletion restriction:** Venues with associated events or bookings cannot be deleted.
3. **Event deletion restriction:** Events with associated bookings cannot be deleted.
4. **Venue name uniqueness:** A unique index enforces that no two venues can share the same name.

Seed Data

The database is seeded with initial data for development and testing:

- **7 Venues:** Grand Ballroom (Johannesburg, 500), Skyline Terrace (Sandton, 200), The Garden Pavilion (Pretoria, 150), Ocean View Hall (Durban, 300), Mountain Lodge (Drakensberg, 80), The Vineyard Estate (Stellenbosch, 250), City Conference Centre (Johannesburg, 1000)
- **8 Events:** Annual Tech Conference, Wedding Reception - Mokoena, Corporate Year-End Gala, Charity Fun Run Launch, Product Launch - Nova Phone, Birthday Celebration - Naidoo, Team Building Retreat, Music Festival Day Pass
- **8 Bookings:** One booking per event, spanning April–October 2026